Banking Customer and Lending Analysis

Business Analysis Report

Project: Banking Customer and Loan Analysis
Tools: Microsoft Excel, SQL Server, SQL Server Management Studio (SSMS)
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1. Executive Summary

This project explores how a bank can grow lending and customer value while managing repayment risk. By analysing customer information, account balances, transaction activity, loan applications and payment records, I identified opportunities to strengthen customer engagement, improve lending decisions and focus risk management where it may be most needed.

The bank has a substantial customer deposit base, with approximately R62.3 million in account balances. Savings accounts contribute approximately R31.1 million, representing half of these balances. The Mobile App is the strongest transaction channel by both value and volume, creating an opportunity to build on existing digital engagement.

Lending is another significant part of the bank's activity. The dataset contains 900 loan records with a combined recorded loan value of approximately R153.7 million. Personal loans represent the largest category by value, while further investigation revealed encouraging growth in vehicle and business loan applications during the first half of the period analysed.

However, growth needs to be considered alongside repayment performance. Approximately 18.2% of recorded loan payments are late or missed. Home loans have the highest late-or-missed payment rate among the loan types examined, while repayment patterns also vary across cities.

The analysis therefore points to an important business priority: the bank should pursue lending opportunities based on demand but use repayment performance to guide where and how it expands. Stronger early-warning measures, more targeted lending strategies and further investigation of higher-risk segments could help support more sustainable growth.

2. Business Problem

Banks need to balance growth with financial risk. Increasing lending can create opportunities to serve more customers and generate interest income, but lending without sufficient attention to repayment behaviour can expose the bank to missed payments and potential credit losses.

At the same time, customer deposits and transaction activity provide opportunities to strengthen existing relationships and introduce relevant products.

The challenge was to understand where the bank's strongest opportunities lie, which lending segments show promising demand, and where repayment outcomes suggest that closer investigation may be necessary.

Rather than simply reporting balances, transaction totals and loan volumes, this analysis aims to connect those results to practical business decisions.

3. Business Questions

The analysis focused on the following questions:

1. Where is customer account value concentrated, and which banking channels are used most?
2. Which loan types and periods show the strongest lending demand?
3. Are changes in lending driven by more applications or by larger loan requests?
4. How do loan approval rates differ across products and over time?
5. Which loan types and cities have higher late-or-missed payment rates?
6. Do the results identify customer segments that warrant further credit-risk investigation?
7. How can the bank pursue lending growth while maintaining proper risk controls?

4. Methodology

Data and preparation

The project used five related tables: Customers, Accounts, Transactions, Loans and LoanPayments. Together, these tables contain information about customer characteristics, account balances, banking activity, loan applications, and repayment outcomes.

The dataset includes 1,500 customers, 2,100 accounts, approximately 15,000 transaction records, 900 loan records and 6,000 loan-payment records. The Transactions table originally included 15 intentionally duplicated rows for cleaning practice.

Tools and analysis

Microsoft Excel was used to inspect the data, support preparation and organise results for reporting and dashboard presentation.

SQL Server and SSMS were used to validate record counts, connect related tables and investigate business questions. The analysis used joins, aggregate functions such as `SUM`, `COUNT` and `AVG`, conditional calculations with `CASE WHEN`, grouping, filtering and date-based comparisons.

The investigation followed a storytelling approach:

Find → Investigate → Interpret → Recommend

Initial results highlighted patterns such as high personal-loan value and differences in repayment performance. Follow-up queries then examined loan-type trends, approval rates and geographic differences to understand the business implications more clearly.

5. Key Findings and Business Implications

Finding 1: Savings balances and digital banking provide opportunities to strengthen customer value

The bank holds approximately R62.3 million in total account balances. Savings accounts contribute approximately R31.1 million, or 50% of the total.

Transaction activity is strongest through the Mobile App, which recorded approximately R6.6 million in transaction value and 4,869 transactions. Online Banking followed with approximately R4.7 million and 3,384 transactions. Branch transactions had the lowest recorded value among the five channels, at approximately R2.0 million.

Why this matters

The savings balance indicates that the bank already has a substantial base of customers holding funds with it. Meanwhile, the Mobile App provides an established channel through which the bank can communicate with customers and make relevant services more accessible.

This creates an opportunity to strengthen existing relationships rather than relying only on acquiring new customers. For example, customers who already maintain savings balances could be offered suitable savings or investment products, while digital channels could support personalised communication and convenient self-service.

The results do not establish that every customer prefers digital banking, but they show that the Mobile App is an important channel in the available transaction data.

Finding 2: Lending demand is substantial, but growth differs across loan products

The recorded loan portfolio contains 900 loan records with a combined value of approximately R153.7 million.

Personal loans represent the largest category, at approximately R51.3 million. Home loans follow at approximately R34.0 million, while vehicle loans account for approximately R25.7 million.

However, the follow-up investigation revealed that demand is not moving in the same direction across all products.

Comparing the first two quarters of each year:

Loan type	2024 applications	2025 applications	2026 applications
Personal	65	55	54
Home	43	49	45
Vehicle	25	29	39
Business	16	36	36
Education	15	22	22

Total applications across these five categories increased from 164 in the first half of 2024 to 191 in 2025 and 196 in 2026.

Vehicle applications increased from 25 to 39 between the first halves of 2024 and 2026. Business applications increased from 16 to 36 and then remained at 36 in the first half of 2026. Personal-loan applications, in contrast, declined from 65 to 54 over the same comparison.

Why this matters

The bank should not assume that every lending category needs the same growth strategy. Vehicle and business lending show increasing application activity in the periods compared, while personal-loan applications have declined.

These patterns could help the bank identify where to investigate customer demand, marketing effectiveness and product suitability. However, application growth alone does not prove that a product is more profitable or that the bank should automatically approve more applications.

Finding 3: More applications do not necessarily mean larger loan requests

The changes in vehicle and business lending become more informative when average application amounts are considered.

For business loans, first-half application volume increased from 16 in 2024 to 36 in 2025 and remained at 36 in 2026. Yet the average requested amount declined from approximately R224,451 in 2024 to R147,052 in 2025 and R141,430 in 2026.

Vehicle lending followed a different pattern. Application volume rose from 25 to 29 and then 39 across the same periods. Average loan amounts increased from approximately R148,109 in 2024 to R226,981 in 2025, before declining to R174,664 in 2026.

Why this matters

Business-loan growth appears to be driven mainly by more applications rather than larger average requests. Vehicle applications are also increasing, although the average requested amount has fluctuated.

This distinction matters for planning. A higher number of applications can increase the workload for credit assessment and processing, even when the average requested amount falls. The bank should therefore monitor application volume, total requested value and average loan amount together rather than relying on one measure of growth.

These figures describe recorded application amounts and should not be treated as confirmed disbursements or realised lending revenue.

Finding 4: Approval rates show differences between products, but growth must still be assessed alongside risk

Across the loan records analysed, overall approval rates differed by loan type:

Loan type	Approval rate
Vehicle	73.42%
Personal	67.38%
Education	66.02%
Home	61.46%
Business	59.21%

The first-half comparisons also revealed changes over time. Business-loan approval rates increased from 56.25% in 2024 to 58.33% in 2025 and 63.89% in 2026. Vehicle-loan approval rates were 72.00%, 75.86% and 71.79%, respectively.

These results suggest that business applications were more likely to be approved in the later comparison period, while vehicle approval rates remained relatively high.

Why this matters

Approval rates help the bank understand how applications progress through its lending process. When considered alongside application volume, they can help identify products where demand and approvals are both increasing.

However, a higher approval rate is not automatically better. Approval decisions must still reflect affordability, creditworthiness and the bank's risk criteria. The bank should evaluate approval trends alongside subsequent repayment outcomes before deciding to expand lending in a particular category.

Finding 5: Repayment performance is the main risk concern

Approximately 81.8% of recorded payments were made on time, while 13.6% were late and 4.7% were missed. This means approximately 18.2% of payments were late or missed.

The follow-up analysis compared this risk measure across loan types:

Loan type	Late or missed payments	Risk rate
Home	251 of 1,250	20.08%
Business	183 of 993	18.43%
Personal	346 of 1,909	18.12%
Education	130 of 724	17.96%
Vehicle	183 of 1,124	16.28%

Home loans recorded the highest late-or-missed payment rate, at 20.08%, while vehicle loans recorded the lowest, at 16.28%.

Personal loans had the largest number of late-or-missed payments, at 346, but they also had the largest number of recorded payments overall. This is why the percentage is important: raw counts alone can make a larger loan category appear riskier simply because it has more payment records.

Why this matters

The results suggest that repayment monitoring should not be applied uniformly without considering differences between loan products. Home loans warrant closer investigation because they have the highest observed risk rate, while the personal-loan portfolio also deserves attention because of its larger number of late-or-missed payments.

The bank should investigate the reasons behind these differences before changing lending criteria. The payment percentages are useful indicators of repayment patterns, but they do not measure the monetary value of losses or establish the probability that an individual borrower will default.

Finding 6: Home-loan repayment risk differs across cities

The geographic investigation examined late-or-missed home-loan payments by customer city.

City	Late or missed payments	Risk rate
Johannesburg	37 of 150	24.67%
Gqeberha	21 of 92	22.83%
Pretoria	20 of 90	22.22%
Bloemfontein	31 of 141	21.99%
East London	25 of 116	21.55%
Cape Town	23 of 108	21.30%
Mbombela	34 of 167	20.36%
Kimberley	29 of 153	18.95%
Polokwane	12 of 85	14.12%
Durban	19 of 148	12.84%

Johannesburg had the highest observed rate, at 24.67%, while Durban had the lowest, at 12.84%. The difference is 11.83 percentage points.

Why this matters

The results show that home-loan repayment outcomes vary geographically. Johannesburg may therefore be a useful starting point for a more focused review of repayment patterns and collection processes.

However, the analysis does not establish that location causes repayment difficulties. Differences could reflect customer characteristics, loan profiles, payment histories or other factors not included in the analysis.

The appropriate response is to investigate these differences further, not to assume that customers in a particular city are inherently higher risk.

Finding 7: Employment status has not explained Johannesburg's higher home-loan risk

The next investigation compared repayment outcomes by employment status among Johannesburg home-loan customers.

Employment status	Late or missed payments	Risk rate
Retired	2 of 8	25.00%
Unemployed	1 of 4	25.00%
Employed	34 of 138	24.64%

The observed rates are very similar, ranging from 24.64% to 25.00%. However, the retired and unemployed groups have very few payment records, making their percentages particularly sensitive to individual payments.

Why this matters

Employment status alone did not reveal a clear difference in repayment risk within this city and loan segment. The results do not support concluding that one employment group is responsible for Johannesburg's higher rate.

The subsequent income investigation also produced results for individual income values rather than well-defined income bands. Those results are not sufficient to establish a reliable relationship between income and repayment risk.

For now, the more defensible conclusion is that geographic differences have been identified, but the underlying customer-level explanation remains uncertain.

Finding 8: The approval-status comparison needs further validation

The final check compared payment outcomes for loans recorded as Approved and Pending.

Loan status	Total payments	Late or missed	Risk rate
Pending	897	168	18.73%
Approved	5,103	925	18.13%

The observed difference is 0.60 percentage points, with Pending loans showing the slightly higher rate.

However, this result needs careful interpretation. A loan marked Pending would not ordinarily be assumed to have progressed through the same process as an approved loan, yet these records have associated payment histories.

Why this matters

The comparison does not establish that approval status meaningfully affects repayment performance. Before using it to inform a business decision, the bank would need to confirm what Pending means in this dataset and why payment records are associated with those loans.

This is an example of why a technically valid SQL result still needs business-context validation before being treated as a reliable insight.

6. Recommendations

Based on the strongest findings, I recommend the following priorities.

1. Strengthen repayment monitoring

Use the overall late-or-missed payment rate of approximately 18.2% as a reason to review repayment monitoring and collections processes.

The bank could identify early signs of payment difficulty and use appropriate reminders or customer-support interventions before problems become more serious. Any intervention should be evaluated against subsequent repayment outcomes.

2. Investigate home-loan repayment risk

Home loans have the highest late-or-missed payment rate among the loan types analysed. The bank should examine the characteristics of these loans and their payment histories to understand what contributes to this pattern.

Johannesburg is a reasonable starting point for that investigation because it has the highest observed city-level rate for home-loan payments. The bank should validate the pattern with additional customer-level evidence before changing credit policies.

3. Evaluate vehicle and business lending opportunities

Vehicle and business applications increased in the first-half comparisons. The bank could investigate whether this demand is translating into suitable approved lending and sustainable repayment outcomes.

For business loans in particular, the increase in application volume alongside a declining average requested amount suggests that both processing capacity and the profile of requests deserve attention.

4. Make lending decisions using demand and risk together

The bank should assess application volume, requested value, approval rates and repayment outcomes together. Strong demand or a high approval rate should not be used alone to justify expanding a loan product.

A balanced view would help the bank distinguish between products with promising demand and products that may require additional risk controls.

5. Build on existing customer and digital engagement

The Mobile App is the strongest transaction channel, and savings accounts hold approximately half of total account balances. The bank could use its established digital channel to improve customer communication and promote suitable savings, investment or lending products.

Offers should be relevant to customers' needs and affordability rather than based solely on account balances or transaction activity.

6. Improve reporting and validation before making larger decisions

The bank should confirm the meaning of loan statuses, review the reliability of customer segmentation and monitor repayment measures over consistent periods. This would help distinguish genuine changes in customer risk from differences in the data or the number of observations.

7. Limitations

Several limitations should be considered when interpreting the results:

- **Recorded loan amounts are not necessarily disbursed amounts.** The analysis describes the loan records and amounts available in the dataset; it does not establish that every recorded amount was paid out.
- **Payment rates describe payment records, not unique borrowers.** One loan can have multiple payments, so a higher payment-level risk rate does not directly tell us how many individual customers are experiencing difficulty.
- **Some segments have small samples.** The retired and unemployed groups in the Johannesburg home-loan analysis have only eight and four payment records, respectively. Their percentages should not be treated as stable estimates.
- **The income investigation was inconclusive.** The results did not provide a reliable comparison across defined income bands, so no firm conclusion about income and repayment risk can be made.
- **Loan status requires clarification.** The Pending category has associated payment records, and its business meaning needs to be confirmed before comparing it directly with Approved.
- **Patterns do not establish causes.** Differences by city or loan type identify where further investigation may be useful, but they do not explain why the differences occur.
- **The dashboard is static.** The current Excel/WPS setup limits interactive exploration, so the findings rely on the SQL analysis and prepared dashboard outputs rather than a fully interactive reporting environment.

8. Next Steps

The next stage is to turn the strongest findings into a clear reporting and decision-making process.

1. **Prioritise the findings:** Lead with repayment risk, differences between loan products and the growth patterns in vehicle and business applications.
2. **Validate the uncertain results:** Clarify Pending loan status and improve the income-segmentation analysis before using to support a business recommendation.
3. **Track relevant measures together:** Monitor application volume, requested loan value, approval rate and late-or-missed payment rate by loan type and period.
4. **Improve the dashboard:** Present the most useful measures clearly, with emphasis on lending trends and repayment risk. Power BI could later make it easier to explore product and geographic differences interactively.
5. **Evaluate recommendations:** Where the bank introduces new campaigns or repayment interventions, compare the outcomes against a suitable baseline to determine whether they are helping.

Conclusion

This analysis shows that the bank has meaningful opportunities to strengthen customer value and develop its lending portfolio, but growth should be guided by repayment performance rather than application volume alone.

The Mobile App and savings balances provide opportunities to build on existing customer relationships. Meanwhile, vehicle and business lending show encouraging application growth in the first-half comparisons. These opportunities need to be considered alongside the repayment findings, particularly the higher observed late-or-missed payment rate for home loans.

The most important lesson from the investigation is that identifying a pattern is only the starting point. Comparing loan types, cities and customer groups helps narrow down where the bank should investigate further, but the evidence must be strong enough to support the proposed action.

By combining demand, approval and repayment measures, validating uncertain results and focusing on the segments that warrant closer review, the bank can make more informed decisions about where to pursue growth and where to strengthen risk management.